

VAN LE

Open to relocation | 206-446-5548 | vanle.vtl2000@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Applied ML scientist building and deploying causal, forecasting, and optimization models from problem framing through production. Independent causal work benchmarking nine estimator configurations (AIPW, IPW, PSM, DiD, g-computation) against RCT ground truth on 14M ad impressions, with uplift and CATE ranking evaluated on Qini. Built a hybrid ML-physics forecasting model over 570K+ observations, and owns a containerized ML system that reconciles ~120K vendor invoices per month behind a benchmarked multi-class classifier at 0.925 macro-F1 tuned under asymmetric misclassification cost. M.S. Computational Data Analytics, Georgia Tech, in progress.

CORE SKILLS

Programming and BI: Python (pandas, scikit-learn, PyTorch), SQL, R, Tableau, Plotly

Machine Learning: Regression, classification, clustering (k-means, DBSCAN, PCA), random forest, XGBoost, RNN, model benchmarking and selection

Causal Inference and Experimentation: A/B testing, RCT analysis, difference-in-differences, propensity score matching, IPW, doubly-robust AIPW, g-computation, statsmodels, EconML

GenAI and LLM Systems: OpenRouter, Claude API, Hermes agent framework, prompt evaluation harnesses, agent guardrails, model explainability via per-decision confidence and abstention

Data and Cloud: AWS (S3, EMR, Athena), BigQuery, PySpark, ETL and pipeline design, data modeling, data quality and integrity validation, Docker, FastAPI, CI/CD, REST API integration (Intuit QuickBooks)

PROFESSIONAL EXPERIENCE

Starpoint Investment

Oct 2024 to Present

Data Scientist

- Deployed a containerized, cloud-native ingestion and decisioning pipeline that automates accounts-payable reconciliation across six platforms (email, DoorDash, Uber, Sysco, Shopify, Amazon), processing **~120K invoices per month** for **10+** client portfolios against QuickBooks via the Intuit API at an **88%** straight-through reconciliation rate, with CI/CD-style gated retraining. Adopted by the accounting team as their primary AP workflow.
- Built and owned the ML classifier behind the AP automation, reaching **0.925 macro-F1** for **~100 GL expense labels** on **10+ years** of client-coded history. Chose per-class thresholds with abstention over global accuracy, since misclassifying an expense as capitalizable equipment changes the client's tax treatment.
- Own the model evaluation framework: a **1,000-case** offline test suite on labeled outcomes gates every prompt, threshold, and model change before deployment, designed to block any regression in the straight-through rate. Built human-in-the-loop guardrails, including automated escalation of low-confidence cases and required manual approval before any high-value financial action.
- Consolidated point-of-sale, e-commerce, inventory, and vendor invoice feeds into BigQuery, unifying **5M+ records** across **10+** client portfolios, and encoded the reconciliation logic tying purchases to inventory movement to recorded sales. Compressed monthly close from **3-5 days to 1 day** and added automated integrity checks for duplicate capture, cross-feed referential integrity, and schema drift.
- Automated **75%** of monthly reporting preparation through LLM-assisted extraction and validation over PDF and Excel vendor documents, saving **15+ hours per month**.

Merck & Co.

May 2023 to Aug 2023

Data Science Intern

- Built a hybrid ML-physics forecasting model (digital twin) that uses an **RNN** to estimate physical parameters from time-series sensor data, feeding a mechanistic simulation to forecast process behavior and optimize control settings for yield and purity. Benchmarked **8** candidate models over **570K+** observations with holdout backtesting on unseen conditions plus residual and sensitivity analysis to map where each model held and where it broke down.
- Engineered PySpark ingestion pipelines on AWS EMR landing manufacturing and instrument data into partitioned S3 tables, lowering ingestion latency **40%**, and authored reusable SQL and Python templates on Athena that cut time-to-data **60%** for four downstream analytics and engineering teams.

Adaptive Biotechnologies

Jun 2022 to Aug 2022

Data Science Intern

- Developed an unsupervised segmentation pipeline over **100K+** single-cell samples with no labeled target: PCA at **40 components** retaining **35.5%** variance, feeding a comparison of k-means against DBSCAN to avoid assuming cluster geometry, with final structure selected on silhouette score and stability under resampling. Revealed previously unrecognized subpopulations that domain scientists validated and adopted as a new research direction, presented to department leadership.

PROJECTS

Causal Inference and Uplift Modeling, Criteo Advertising Data [GitHub](#)

- Recovered a known causal effect from deliberately broken data: **14M** ad impressions with confounding injected by a prognostic-score selection rule at **0.947** holdout AUC, pushing the naive conversion ATE to **3.4x** the RCT ground truth. Doubly-robust AIPW reduced bias from **+237% to +4.6%**, while IPW reached textbook covariate balance at all |SMD| below 0.05 yet remained **92% biased** as its effective control sample collapsed to **192K of 1.06M**. Benchmarked nine estimator configurations (PSM, IPW, AIPW, g-computation, DiD under satisfied and violated parallel trends) against placebo tests, propensity trimming, and Rosenbaum bounds.
- Ranked audiences by predicted CATE on a cluster-split holdout of **3.6M** rows with no user appearing on both sides: a transformed-outcome learner reached **Qini 0.659** and **6.6x** uplift in the top decile, while a plain response model outranked both causal learners at **Qini 0.813**, establishing that uplift modeling must be justified against that baseline. Served the selected model behind FastAPI in Docker with gated retraining on a holdout-Qini floor and PSI drift monitoring.

Constrained Budget Optimization for Marketplace Campaigns [GitHub](#)

- Formulated and evaluated a fixed-budget reallocation policy (constrained optimization): isolated low-efficiency campaign segments consuming **34.6%** of budget while producing only **22.9%** of approved conversions across **\$20.3K** spend, then modeled a reallocation policy under fixed-budget constraint, projecting **+148** incremental approved conversions and CPA improvement from **\$58.53 to \$50.98**.

EDUCATION

Georgia Institute of Technology, M.S. Computational Data Analytics

Expected Dec 2028

SUNY Buffalo, B.A. Computational and Applied Mathematics

Aug 2022 to Dec 2023